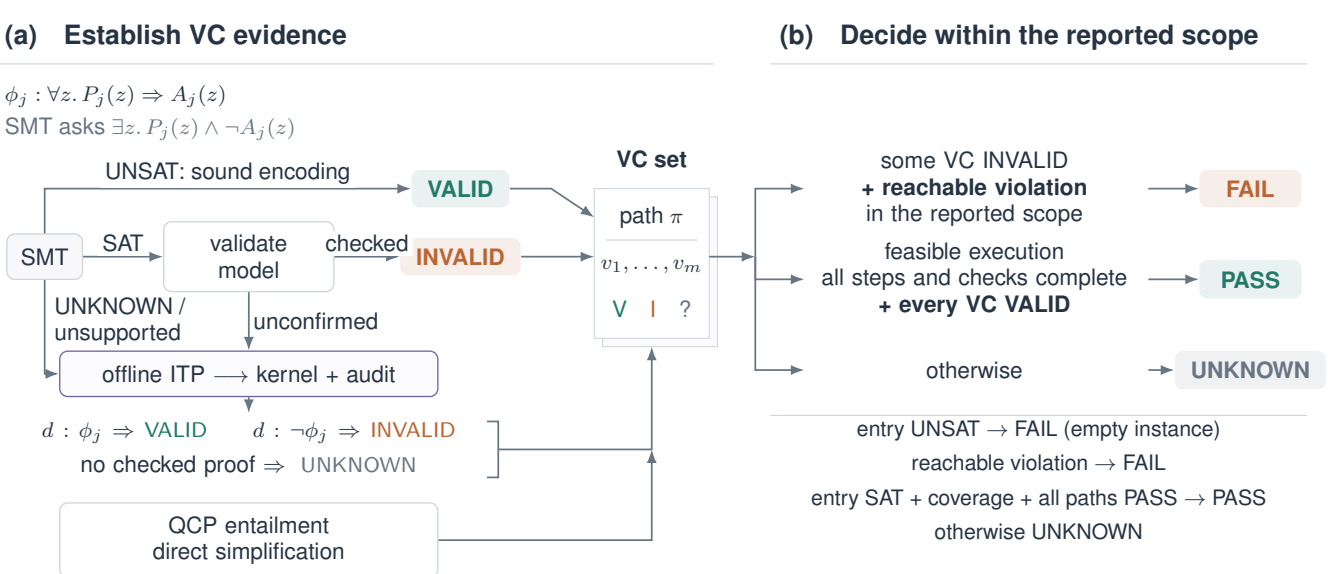

(a) Establish VC evidence
$\phi_j : \forall z.\, P_j(z) \Rightarrow A_j(z)$
SMT asks $\exists z.\, P_j(z) \wedge \neg A_j(z)$
UNSAT: sound encoding
VALID
SMT
SAT
validate model
checked
INVALID
UNKNOWN / unsupported
unconfirmed
offline ITP ⟶ kernel + audit
$d : \phi_j \Rightarrow$ VALID
$d : \neg\phi_j \Rightarrow$ INVALID
no checked proof ⇒ UNKNOWN
QCP entailment
direct simplification
VC set
path π
$v_1, \ldots, v_m$
V I ?
(b) Decide within the reported scope
some VC INVALID
+ reachable violation
in the reported scope
FAIL
feasible execution
all steps and checks complete
+ every VC VALID
PASS
otherwise
UNKNOWN
entry UNSAT → FAIL (empty instance)
reachable violation → FAIL
entry SAT + coverage + all paths PASS → PASS
otherwise UNKNOWN